

4. 利用第 3 题的结果.

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6. 记 $f(x, y) = \int_x^y f(t)dt$, 则 $f(x, y) = -f(y, x)$. 交换积分变元 x, y , 再用 Fubini 定理, 可得

$$\begin{aligned}\int_a^b dx \int_a^b f(x, y)dy &= \int_a^b dy \int_a^b f(y, x)dx \\ &= - \int_a^b dy \int_a^b f(x, y)dx = - \int_a^b dx \int_a^b f(x, y)dy.\end{aligned}$$

另一证法:

$$\begin{aligned}\int_a^b dx \int_a^b dy \int_a^b f(t)dt &= (b-a)^2 \int_a^b f(t)dt; \\ \int_a^b dx \int_a^b dy \int_y^b f(t)dt &= (b-a) \int_a^b (t-a)f(t)dt; \\ \int_a^b dx \int_a^b dy \int_a^x f(t)dt &= (b-a) \int_a^b (b-t)f(t)dt;\end{aligned}$$

所以

$$\begin{aligned}\int_a^b dx \int_a^b dy \int_x^y f(t)dt &= \int_a^b dx \int_a^b dy \left(\int_a^b f(t)dt - \int_a^x f(t)dt - \int_y^b f(t)dt \right) \\ &= \int_a^b f(t)[(b-a)^2 - (b-a)(t-a) - (b-a)(b-t)]dt = 0.\end{aligned}$$

$$8. \int_0^1 \int_0^1 f(x, y)dx dy = 1.$$

9. 用反证法.

10. 第 10 题是第 11 题的特例. 用 Tonelli 定理直接证明或用第 11 题的结论.

$$11. \varphi(x) = \int_{E(g \geq y)} f(x)dx = \int_E f(x)\chi_{E(g \geq y)}(x)dx, \varphi(y) \text{ 是非负可测函}$$

数的积分, 所以存在, 且有